26900078 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOTA04 - PLANT-BASED NUTRACEUTICALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Listout any two important microbial sources of nutraceuticals. [K] / [CO1]
2. Mention two examples of dietary supplements. [K] / [CO1]
3. Explain one characteristic property of saponins [U] / [CO1]
4. Mention two traditional uses of Ocimum sanctum. [U] / [CO2]
5. Identify two nutrients commonly obtained from plant foods. [U] / [CO2]
6. Outline the importance of tribal plant knowledge in nutraceutical research. [A] / [CO2]
7. Explain how dietary fibre supports health. [U] / [CO3]
8. Name the plant part of fenugreek commonly used for its nutraceutical properties. [K] / [CO3]
9. Explain the meaning of a side effect. [U] / [CO3]
10. Define fluorescence [U] / [CO4]

11. Expand TKDL. [U] / [CO4]

12. Explain the principle of Soxhlet extraction [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **8** questions

Each question carries **5** marks

13. Explain the characteristics and examples of natural nutraceuticals. [U] / [CO1]

14. Discuss the role of nutraceuticals in supporting normal immune function. [U] / [CO1]

15. Analyse the role of fruit and leaf powders as functional ingredients. [An] / [CO1]

16. How can ethnobotanical knowledge be applied in nutraceutical discovery? [A] / [CO2]

17. Describe the ethical practices that should be followed during ethnobotanical fieldwork. [U] / [CO2]

18. Suggest how plant-based nutraceuticals can be used for immune support. [A] / [CO3]

19. Analyse the potential role of Ashwagandha in supporting brain health and memory. [An] / [CO3]

20. Discuss the benefits and significance of plant-based nutraceuticals [U] / [CO3]

21. Analyse the importance of correct dosage in ensuring the safety and efficacy of nutraceutical products. [An] / [CO4]

22. Analyse the impact of safety approval and responsible marketing on nutraceutical safety and consumer confidence. [An] / [CO4]

[5x8 = 40]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

23. Discuss the importance of traditional knowledge in identifying nutritionally and medicinally useful plants and explain its role as a starting point for plant-based nutraceutical discovery [U] / [CO2]

24. Explain the selection and preparation of plant materials for nutraceutical extraction and describe the principle, procedure, factors and advantages and limitations of solvent extraction. [U] / [CO4]

[10x1 = 10]